Title: Moving Average Crossover Trading Strategy

1. Start.

2. Load historical daily stock data containing Open, High, Low, Close and Volume (OHLCV).

3. Validate the data and check for missing values and survivorship bias.

4. Divide the data chronologically into 70% development data and 30% final test data.

5. Calculate SMA 20, SMA 50, SMA 200 and ATR 14.

6. Check whether sufficient historical data is available. If not, wait for more data.

7. Check whether a position is already open.

8. If no position is open, check whether SMA 20 crosses above SMA 50.

9. If it does, check whether the closing price is above SMA 200.

10. If both conditions are satisfied, generate a buy signal after market close.

11. Buy at the next trading day's opening price. Apply transaction costs and calculate the stop-loss, profit target and position quantity using the predefined risk limit.

12. If the quantity is invalid or unaffordable, skip the trade. Otherwise, record the position.

13. If a position is open, check the stop-loss and profit target. If either is triggered, exit using the predefined conservative execution rules.

14. If neither is triggered, check whether SMA 20 crosses below SMA 50. If it does, exit at the next available execution price. Otherwise, hold the position.

15. Update cash, holdings, transaction costs and portfolio value.

16. Repeat the trading process for each remaining trading day.

17. Freeze the strategy rules and evaluate the untouched final 30% of the data.

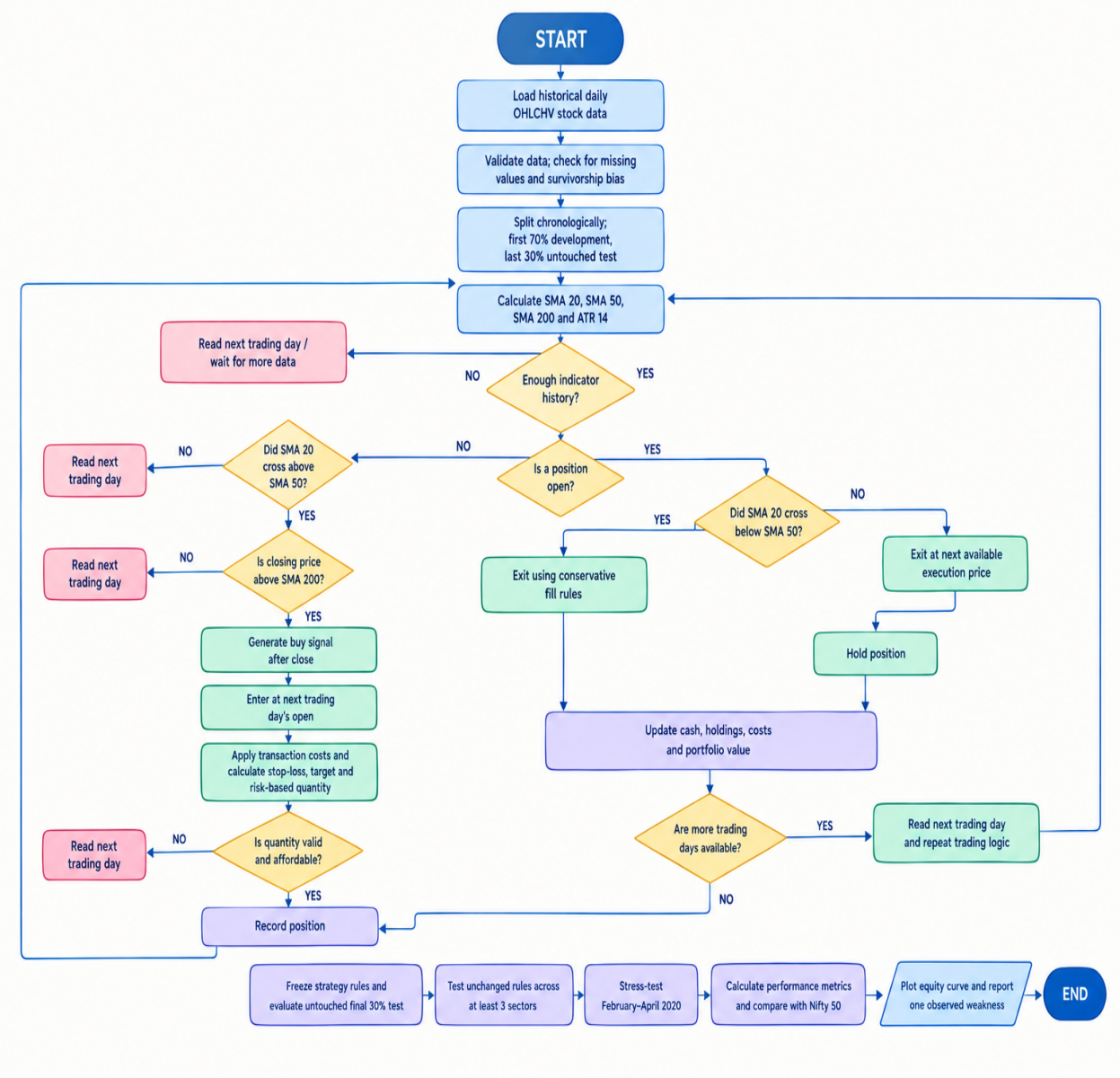
18. Test the unchanged strategy across at least three sectors.

19. Stress-test the strategy using February–April 2020 data.

20. Calculate performance metrics, compare the results with the Nifty 50, plot the equity curve and report one observed weakness.

21. End.

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START
Load historical daily OHLCHV stock data
Validate data; check for missing values and survivorship bias
Split chronologically; first 70% development, last 30% untouched test
Calculate SMA 20, SMA 50, SMA 200 and ATR 14
Enough indicator history?
NO
YES
Read next trading day / wait for more data
Is a position open?
NO
YES
Did SMA 20 cross above SMA 50?
NO
YES
Read next trading day
Is closing price above SMA 200?
NO
YES
Read next trading day
Generate buy signal after close
Enter at next trading day's open
Apply transaction costs and calculate stop-loss, target and risk-based quantity
Is quantity valid and affordable?
NO
YES
Read next trading day
Record position
Did SMA 20 cross below SMA 50?
YES
NO
Exit using conservative fill rules
Exit at next available execution price
Hold position
Update cash, holdings, costs and portfolio value
Are more trading days available?
YES
NO
Read next trading day and repeat trading logic
Freeze strategy rules and evaluate untouched final 30% test
Test unchanged rules across at least 3 sectors
Stress-test February–April 2020
Calculate performance metrics and compare with Nifty 50
Plot equity curve and report one observed weakness
END